Variable-depth inverse reduction and the first two decoder layers

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For a positive odd integer x, put

$$U_+(x) := \frac{3x+1}{2^{v_2(3x+1)}}. \tag{1}$$

For a positive exponent word $w = (a_1, \dots, a_m)$, define

$$\mathrm{B}(w) := \sum_{i=1}^{m} a_i, \qquad \mathrm{C}(w) := \sum_{i=1}^{m} 3^{m-i} 2^{a_1+\cdots+a_{i-1}}, \tag{2}$$

and its shifted carry by

$$\mathrm{D}(w) := \mathrm{C}(w) + 2^{\mathrm{B}(w)} - 3^m. \tag{3}$$

Thus an exact path with word w from x to y satisfies

$$2^{\mathrm{B}(w)}(y+1) = 3^m(x+1) + \mathrm{D}(w). \tag{4}$$

We use the record atoms

$$A := 1, \qquad B := 12, \qquad C := 1113, \qquad D := 1122. \tag{5}$$

1 A sharp prefix interval

Lemma 1 (shifted-carry interval)**.** *Let $E = (e_1, \dots, e_h)$ be a positive exponent word and write*

$$\mathrm{B}(E) = h + t, \qquad t \geq 0. \tag{6}$$

Then

$$0 \leq \mathrm{D}(E) \leq 3^h(2^t - 1). \tag{7}$$

Equality on the right occurs if and only if $t = 0$, in which case $E = 1^h$ and $\mathrm{D}(E) = 0$.

If E is an exact path from a positive odd z to a positive odd r, put

$$\mathcal{A}_h(r) := \left(\frac{2}{3}\right)^h (r+1). \tag{8}$$

Then

$$\boxed{2^t\big(\mathcal{A}_h(r) - 1\big) \le z \le 2^t \mathcal{A}_h(r) - 1.} \tag{9}$$

The lower bound in (9) is attained if and only if $t = 0$.

Proof. Expanding (3) one letter at a time gives

$$\mathrm{D}(E) = \sum_{i=1}^{h} 3^{h-i} 2^{e_1+\cdots+e_{i-1}}(2^{e_i} - 2) \ge 0. \tag{10}$$

Moreover,

$$\frac{\mathrm{C}(E)}{2^{\mathrm{B}(E)}} = \sum_{i=1}^{h} \frac{3^{h-i}}{2^{e_i+\cdots+e_h}} \le \sum_{j=0}^{h-1} \frac{3^j}{2^{j+1}} = \left(\frac{3}{2}\right)^h - 1. \tag{11}$$

The inequality in (11) is an equality precisely when every $e_i = 1$. Substitution in (3) proves (7), including its equality assertion.

Equation (4) for E is

$$z + 1 = 2^t \mathcal{A}_h(r) - \frac{\mathrm{D}(E)}{3^h}. \tag{12}$$

The two bounds in (7) now give (9). Equality at the lower edge forces equality in (7), hence $t = 0$; the converse follows from (10). □

2 The exact variable-depth factorization

We use the arbitrary-prefix same-depth inverse theorem: if

$$W = PU, \qquad P \in \{B, C, D\}, \qquad U \in \{A, B, C, D\}^*, \tag{13}$$

is exact from q to y and has raw length L, then its least positive odd L-step predecessor is

$$q_* := \begin{cases} q, & P = B \text{ or } D, \\ (q-1)/2, & P = C. \end{cases} \tag{14}$$

It is unique; in the second case its word is $1121U$.

The same theorem gives a useful refinement. If F is any exact length-L word from r to y, write

$$\delta := \mathrm{B}(F) - \mathrm{B}(W), \qquad r = 2^{\delta} q + c. \tag{15}$$

Its distortion proof gives $\delta \geq -1$ and $-1 \leq c/2^{\delta} \leq 5$. At defect -1 one has exactly the second case of (14). At defect zero, parity gives $c \in \{0, 2, 4\}$. In fact $c = 4$ is impossible: comparison of the two shifted equations gives

$$\mathrm{D}(F) = \mathrm{D}(W) - c3^{L}, \tag{16}$$

where $0 \leq \mathrm{D}(F)$ and $\mathrm{D}(W) < 4\,3^{L}$. Consequently every F lies in exactly one of the following channels:

defect	possible start	equality data
-1	$(q-1)/2$	$P = C, \quad F = 1121U,$
0	q or $q+2$	$r = q \iff F = W,$
$\delta \geq 1$	$r \geq 2^{\delta}(q-1)$	.

(17)

Theorem 2 (variable-depth channel reduction)**.** *Let $W = PU$ be as in (13), and let V be any exact word of length $L + h$ from a positive odd integer z to y, where $h \geq 0$. Split it uniquely as*

$$V = EF, \qquad |E| = h, \qquad |F| = L, \tag{18}$$

and put

$$r := U_{+}^{h}(z), \qquad t := \mathrm{B}(E) - h. \tag{19}$$

Then $t \geq 0$, the suffix (F, r) belongs to exactly one of the three channels (17), and

$$\boxed{2^{t}\big(\mathcal{A}_h(r) - 1\big) \leq z \leq 2^{t}\mathcal{A}_h(r) - 1.} \tag{20}$$

In particular,

$$\boxed{z \geq 2^{t}\big(\mathcal{A}_h(q_*) - 1\big).} \tag{21}$$

Put $b_* := v_3(q_* + 1)$. For every $0 \leq h \leq b_*$, the unique least positive odd $(L + h)$-step predecessor of y is

$$\boxed{z_h + 1 = \left(\frac{2}{3}\right)^{h} (q_* + 1).} \tag{22}$$

Its word is

$$\boxed{\begin{cases} 1^{h}W, & P = B \text{ or } D, \\ 1^{h}1121U, & P = C. \end{cases}} \tag{23}$$

Proof. The split (18) is made after the first h exact steps of V; hence E is exact from z to r and F is exact from r to y. The channel statement follows from (17), while (20) is Lemma 1 applied to E. Since every channel has $r \geq q_*$, monotonicity in r proves (21).

If $h \leq b_*$, the right side of (22) is an even positive integer. Starting from z_h, the first h exponent-one steps are exact and end at q_*. Appending the unique minimum suffix in (14) gives (23). This path has $t = 0$ and $r = q_*$, so it attains (21). Equality in (21) forces equality both in $r \geq q_*$ and in Lemma 1; hence $r = q_*$, $t = 0$, $E = 1^h$, and the suffix is the unique word in (14). This proves the asserted uniqueness. □

Corollary 3 (necessary low-excess condition for a new descent)**.** *In the notation of Theorem 2, suppose $\mathcal{A}_h(q_*) > 1$. If $z < N$ for some positive threshold N, then necessarily*

$$\boxed{2^t < \frac{N}{\mathcal{A}_h(q_*) - 1}.} \tag{24}$$

In particular, a start below q_* requires

$$t < h \log_2\left(\frac{3}{2}\right) + \log_2\left(\frac{q_*}{q_* + 1 - (3/2)^h}\right). \tag{25}$$

Thus the remaining prefix problem is a low-excess inverse problem, not an unrestricted search over exponent words.

Proof. Equation (24) is the contrapositive of (21). Taking $N = q_*$ and factoring

$$\mathcal{A}_h(q_*) - 1 = \left(\frac{2}{3}\right)^h \left(q_* + 1 - (3/2)^h\right)$$

gives (25). □

3 The excess-zero and excess-one layers

Lemma 4 (first two decoder layers at a fixed suffix start)**.** *Let r be a positive odd integer and put*

$$b := v_3(r+1), \qquad s := \frac{r+1}{3^b}, \qquad N_1 := 2^{b+1}s - 1, \qquad b_1 := v_3(N_1). \tag{26}$$

Then the following classifications are exact.

1. *An exact h-step predecessor word of excess* $\mathrm{B}(E) - h = 0$ *exists if and only if* $0 \le h \le b$. *It is unique:*

$$E = 1^h, \qquad z + 1 = \left(\frac{2}{3}\right)^h (r + 1). \tag{27}$$

2. *Let* $h > b$ *and write* $j := h - b$. *An exact h-step predecessor word of excess* $\mathrm{B}(E) - h = 1$ *exists if and only if*

$$1 \le j \le b_1. \tag{28}$$

It is unique:

$$\boxed{E = 1^{j-1} 2\, 1^b, \qquad z + 1 = \left(\frac{2}{3}\right)^j N_1.} \tag{29}$$

Proof. Excess zero forces every exponent to equal one. One inverse step of exponent one replaces $x + 1$ by $2(x + 1)/3$. Exactly b such steps can be reversed from r, proving (27).

An excess-one word has a unique exponent two. If it occurs in position j from the start, then (10) gives

$$\mathrm{D}(E) = 2^j 3^{h-j}. \tag{30}$$

The integrality condition in (4) is therefore

$$2^{h+1-j}(r + 1) \equiv 3^{h-j} \pmod{3^h}. \tag{31}$$

Because $h > b$, comparison of 3-adic valuations in (31) forces $h - j = b$. After division by 3^b, the remaining congruence is

$$2^{b+1} s \equiv 1 \pmod{3^j}, \tag{32}$$

which is equivalent to $j \le b_1$. This proves necessity and the uniqueness of the proposed word.

For sufficiency, reverse the b exponent-one steps nearest r. The shifted state is then $2^b s$. Congruence (32) for $j \ge 1$ makes the next inverse exponent exactly two, and its predecessor has shifted value $2N_1/3$. The remaining $j - 1$ inverse exponent-one steps are exact because $3^j \mid N_1$. Their final shifted value is $(2/3)^j N_1$, proving (29). □

Corollary 5 (fixed-seed thresholds and the next exact gate)**.** *Let n be the globally least positive odd integer with an unbounded accelerated orbit, and suppose that the orbit of r is unbounded. Then every word in Lemma 4 has start at least n. Consequently*

$$r + 1 \geq (n + 1) \left(\frac{3}{2}\right)^h \qquad (0 \leq h \leq b), \tag{33}$$

and

$$N_1 \geq (n + 1) \left(\frac{3}{2}\right)^j \qquad (1 \leq j \leq b_1). \tag{34}$$

Apply this with $r = q_*$ in Theorem 2. The minimum same-depth channel is completely exhausted through its excess-zero ladder and its entire post-ladder excess-one layer. After the last depth covered by (27) and (29), every prefix in that channel has excess at least two. Hence a proof by a longer inverse must now do at least one of the following:

1. *control the nonminimal suffix channels $r = q$, $r = q + 2$, and $r \geq 2^\delta(q - 1)$ from (17); or*
2. *control excess at least two beyond the two explicit decoder layers of the minimum channel.*

Any candidate start below n must additionally satisfy the sharp necessary inequality (24) with $N = n$.

Proof. Every positive predecessor of r also has an unbounded forward orbit, so its minimality gives $z \geq n$. Substitution of (27) and (29) proves (33) and (34). Lemma 4 proves the exhaustion assertion, Theorem 2 gives the three suffix channels, and Corollary 3 gives the final necessary inequality. □